An Interactive Framework to Develop and Align Business Process Models

Dorob Wali Ahmad
Northern Health
Prince George BC, Canada
dorob.waliahmad@northernhealth.ca

Waqar Haque
University of Northern British Columbia
Prince George, BC, Canada
waqar.haque@unbc.ca

***Abstract*— In the past few decades, the recognition and adoption of Business Process Management (BPM) has increased at a phenomenal pace. Organizations are increasingly devoting resources towards development and use of BPM techniques and technologies to model, analyze, improve and implement business processes. However, the existing procedures for collecting information to create business process models generally lead to misunderstanding or ambiguity between model and domain experts. We propose a framework to automate collection of such information and build models directly from users' inputs captured through interactive web-forms. The framework also allows users to align processes with strategic business objectives, critical success factors, key performance indicators and functions. Further, processes can be tagged with appropriate maturity level, types and tiers. A dashboard then provides real-time reports which help decision makers to monitor organization's performance, make better decisions, and standardize/optimize processes across the organization.**

Keywords—Business Process Management, Business Process Model, Strategic Alignment

I. INTRODUCTION

A business process is a chain of activities, events, and decisions involving several actors and objects in order to complete organizational tasks [1]. The business process model is a diagrammatic representation or blue print of the business process which presents itself for analysis and optimization in order to add value to the organization and its customers. These processes go through an iterative lifecycle for continuous improvement, ensure consistent outcomes and provide various benefits such as reduced costs, faster execution time and fewer errors. To capture the breadth, Business Process Management (BPM) has been appropriately defined as "a discipline involving any combination of modeling, automation, execution, control, measurement, and optimization of business activity flow in applicable combination to support enterprise goals, spanning organizational and system boundaries, and involving employees, customers, and partners within and beyond the enterprise boundaries" [2]. Importantly, BPM makes the organization's workflow effective, efficient and adaptable to an ever-changing business environment. BPM is used in various areas of an organization including human resources, finance and purchasing, management of customer relationships, sales and marketing, research and development.

Organizations are increasingly moving their business processes online making it easier to access and facilitate extraction of business logic from the business process models. While creating such models is envisioned to be an asset, it could easily become a liability. For example, if an organization opts to model each of its processes, this could yield an unmanageable number of model and incur a heavy burden in terms of money, time, human and capital resources. Similarly, when business processes are not aligned to the organization's Strategic Business Objectives (SBOs), Critical Success Factors (CSFs), Key Performance Indicator (KPIs), and functions, it is futile to improve such processes as they do not offer value to the organization. Additionally, business processes may become a liability when they are not standardized and, therefore, may include redundancies.

There has been an increasing demand for techniques to create business process models efficiently. Several methods have been proposed but the existing procedures for collecting relevant information almost always lead to misunderstanding or ambiguity between model and domain experts. In this paper, we present a framework aimed at generating effective and efficient, strategically aligned, business process models using input collected through asynchronous collaboration from all involved users. Organizations can then put the desired effort into design, implementation, execution, monitoring and evaluation of the generated models.

II. RELATED WORK

BPM enhances collaboration and communication among IT and other business areas within an organization in the process of development, implementation and optimization of business processes. Over decades, organizations have used techniques such as business process reengineering and Six Sigma to optimize their operations [3]. Lusk et al. [4] state that BPM and related tools evolved as results of customization, innovation, increased one-to-one customer focus, and business growth that has been caused by consolidation. Due to the increased adoption of business process management, extensive research has been done in this area. We have divided this into three categories: generation, extraction of information, and improvement of business process models.

A. Generating Business Process Models

Redding et al. [5] proposed an approach to generate business process models from object behavior models using a three-step

978-1-7281-5953-9/19/$31.00 ©2019 IEEE

transformation process. This approach covers the control flow of objects but does not cover resource allocation and data flow. An automated approach towards generating BPMN (Business Process Model and Notation) models from document description was proposed by Friedrich et al. [6]. NPL (Natural Language Processing) and BPM have been combined to achieve this goal. A set of 47 text-model pairs from industry and textbooks were used, and the generated models were on average 77% correct when compared with human created models. During the creation of these models, it was assumed that the process descriptions were sequential, do not contain questions, and the description documents have less irrelevancy information about the processes. Other related work includes visualization of use-case sets as BPMN processes [7], generation of business process model for a set of given reference object life cycle [8], and construction of simple artifact-based models [9].

A common shortcoming of the existing approaches and methods is that the domain experts are not involved in the process. In general, the model experts build these models using software and tools like iGrafx® Flowcharter [10], Visual Paradigm [11], Oracle BPM Suite [12], Bizagi BPM Suite [13], IBM BPM [14], Bonita BPM [15], etc. However, processes are not aligned with strategic business objectives of the organization, and do not have the capability to assign process tier and process type to business process models.

B. *Extracting Information from Business Process Models*

Here, we present a reverse perspective, that is, the approaches to extract information from existing models. Meyer and Weske [16] presented an approach to extract data objects and their status from business process models. This approach is very useful when organizations want to move from documentation of processes to their execution. It also assists the organization to validate process models with the extracted data objects. Tomas et al. [17] proposed a method to extract vocabulary from the business process models. This algorithm is useful in a large BPM project where multiple models are involved and need to be integrated. One of the issues in this algorithm is the process of manually identifying verbs, nouns and adjectives in the first stage, and then forming business expressions. Leopold et al. [18] developed a method to manually create natural language text from BPMN process models selected from several areas. The generated text can assist the domain expert to better understand the model and could also be used for training purposes. The results indicated that 66 percent of generated text was correct, that is, similar to that created by human. However, these results may be biased because the participants were well conversant with BPMN.

C. *Improving Business Process Models*

An automated layout algorithm was implemented which takes BPMN models in eRDF format as input and improves the layout of the models, but not description of the process [19]. Another technique which automatically detects the difference between business process models and textual descriptions was presented by Van Der Aa et al. [20]. The proposed approach uses natural language processing methods to identify inconsistencies between business process models and textual description with a very low number of false results. Wetzstein et al. [21] explained how ontology and semantic technology could be used to improve automation of the BPM lifecycle. Dijkman et al. [22] presented three similarity metrics to retrieve process models from a repository that closely matches the given process model. The proposed metrics were tested with two different datasets and concluded that the results which were finding similarity between given model and those in the repository were better than the ones obtained from text-based search engine. Pittke et al. [23] proposed an approach that recommends the most relevant name for the elements within process models. An evaluation was conducted using test data, evaluation design, evaluation metrics and prototype implementation.

Organizations create workshops and train process experts with new techniques in order to improve the business process lifecycle. The existing approaches make several unrealistic assumptions and suffer from other shortcomings which establishes the need for a more effective approach.

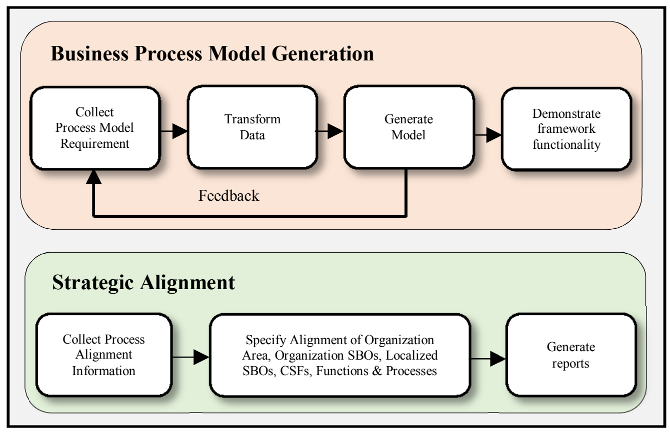


Fig 1. High-level architecture design

III. The Proposed Framework

In this section, we present our proposed framework which is designed to create business process models more efficiently and effectively, and align processes with strategic business objectives of the organization.

A. *Architecture overview*

The high-level architecture comprises of two main components: Business Process Model Generation and Strategic Alignment (Fig 1). To generate the model, users provide process requirements process alignment information through structured forms. The alignment spans organizational SBOs, localized SBOs, Critical Success Factors, functions and processes. These requirements are transformed into XPDL files which are imported into Bizagi Modeler [13] for a visual representation of the model. The users have the ability to edit process requirements and regenerate the model, as necessary. The comprehensive reports generated from the user input contain aggregated information with drill down and drill through capabilities.

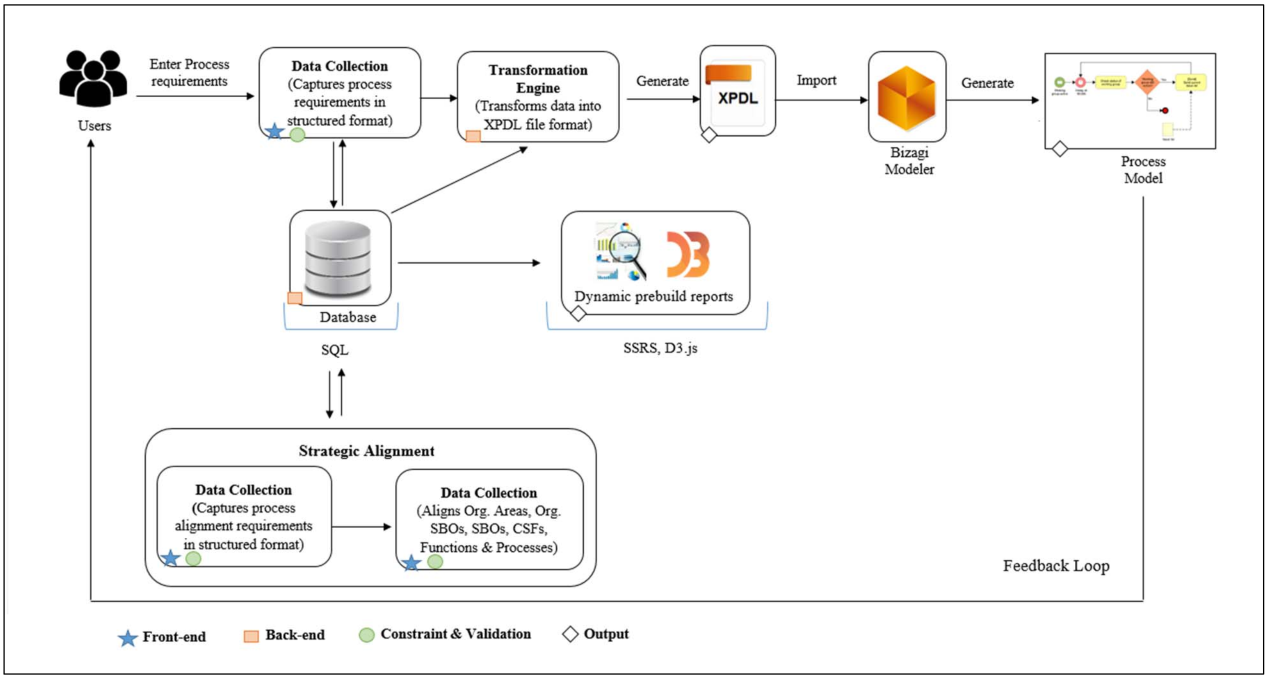


Fig 2. Framework prototype

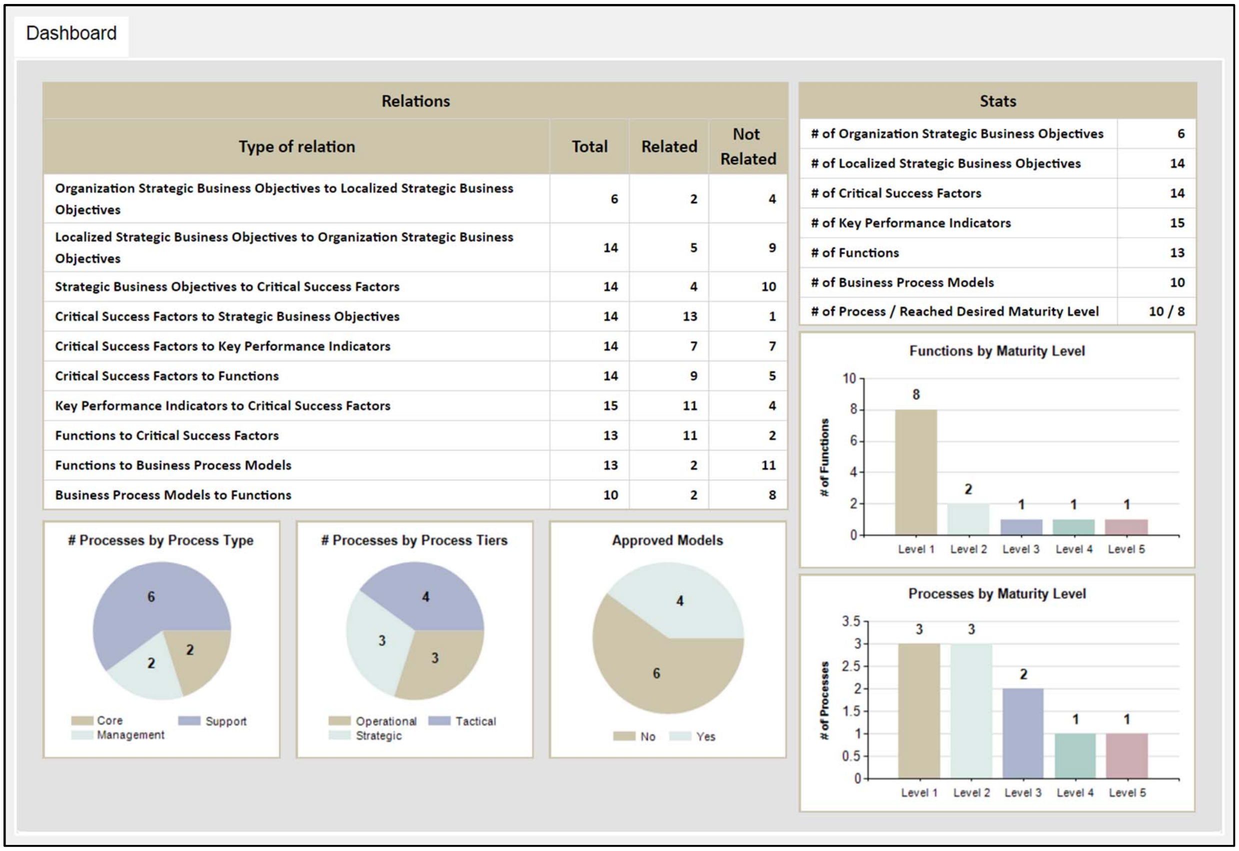
B. *Framework prototype*

Our proposed framework consists of a front-end and back-end (Fig 2). The purpose of front-end component is to provide an intuitive/user-friendly interface for submitting process requirements. The main components of this interface are data collection, strategic alignments, and dynamic prebuilt reports, which allow the users to easily interact with the framework. Data integrity constraints are also enforced within this component. The structured forms to collect process requirements are built with ASP.NET, C#, Hyper Text Markup Language (HTML), JavaScript, Cascading Style Sheets (CSS), and Server Scripting. The collected information is stored in Microsoft SQL Server database using SQL (Structured Query Language) and various reports are generated using this information. The top-level dashboard (Fig 3) consists of relevant charts which provide real time information about the business process models and their relationships. The reporting is further augmented by drill-down and drill-through reports on specific KPIs. The reports are created using SSRS (SQL Server Reporting Services), D3.js (a JavaScript library), Hypertext Markup Language (HTML5), and Cascading Style Sheets (CSS). The data is converted into JavaScript Object Notation (JSON) file format for reports which are built in D3.js.

The back-end components are hidden from users and are built using C#. These include data storage, transformation engine, validation component to validate the accuracy of data for redundancy, and user authentication. The constraints and validation allow to remove data redundancy, improve accuracy and ensure BPMN compliance. The activities, events and decisions that occur within a process are automatically arranged in a sequential order and stored in the database. This data is then transformed into an XPDL file which is imported into the Bizagi Modeler [13] to create the business process model. The users add feedback for the generated model(s) and can also submit change(s) for the processes if needed. The steps involved in transformation engine are shown in Fig 4.

a) Model Layout: When elements are created, the information on their position is unknown. Before creating pool and lanes, the height of each lane is calculated based on the comment (text annotation), data object (attached file to activity) and antecedent flow object within that lane. While building the XPDL file for the model, the position (x-coordinate and y-coordinate) of the element is automatically calculated by our proposed framework. The y-coordinate for a flow object is based on the lane that the element belongs to and also the number of elements that follow the specific antecedent element. The position of the flow object changes if it has more than one antecedent flow objects. The x-coordinate, on the other hand is determined by the flow of an element and number of antecedent elements. The model layout aligns and arranges each component (pools, lanes, flow objects & gateways) of the model automatically.

b) Feedback loop: The framework facilitates editing and updating through the feedback loop. This prevents the users from going through the entire process of entering information when changes occur in business processes. The edit and delete functionality on each data collection page allows the user to modify or remove information for the process(es); this feature is available across all web-forms. For instance, using the edit functionality for manage backup and recovery process (Fig 5), process type can be changed from support to management, or maturity level can be changed from level 2 to level 4.



Type of relation	Total	Related	Not Related
Organization Strategic Business Objectives to Localized Strategic Business Objectives	6	2	4
Localized Strategic Business Objectives to Organization Strategic Business Objectives	14	5	9
Strategic Business Objectives to Critical Success Factors	14	4	10
Critical Success Factors to Strategic Business Objectives	14	13	1
Critical Success Factors to Key Performance Indicators	14	7	7
Critical Success Factors to Functions	14	9	5
Key Performance Indicators to Critical Success Factors	15	11	4
Functions to Critical Success Factors	13	11	2
Functions to Business Process Models	13	2	11
Business Process Models to Functions	10	2	8

Stats	
# of Organization Strategic Business Objectives	6
# of Localized Strategic Business Objectives	14
# of Critical Success Factors	14
# of Key Performance Indicators	15
# of Functions	13
# of Business Process Models	10
# of Process / Reached Desired Maturity Level	10 / 8

Fig 3. Main Dashboard

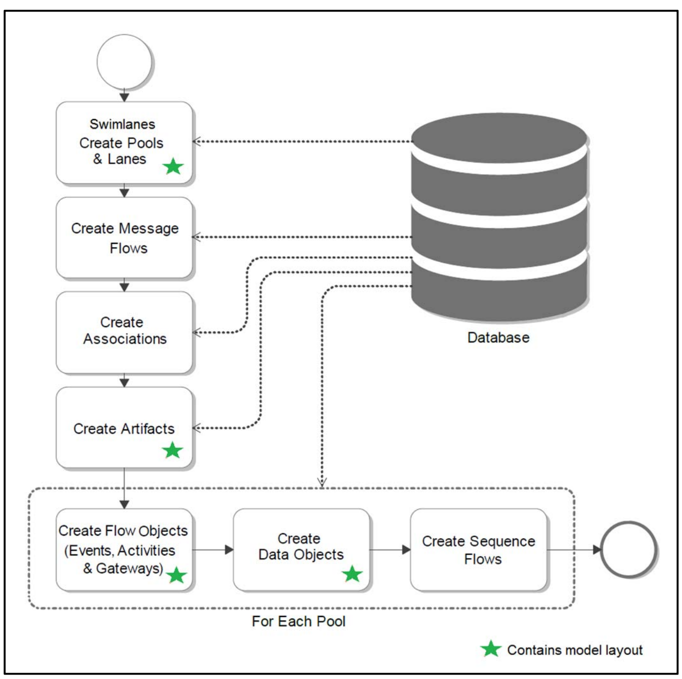


Fig 4.Steps for data transformation

c) Constraints and validation: The constraints and validation allow removal of data redundancy, improve accuracy, and make the models BPMN compliant. Several constraints and validations are embedded in our proposed framework. The *decision constraint* forces the user to add the descendant tasks that follow the decision in order to make it BPMN compliant. For example, at least two pathways and descriptions must be provided for any decision. The *activity type constraint* requires the users to identify the type of tasks, for example, whether the task is performed manually, automatically, with the help of software, or if a business rule is attached to the task. Several fields are also designated as *required* to ensure that no mandatory information is missed. For example, the framework does not allow the user to submit the form without providing task description. Finally, *prepopulated information* (for example, roles and departments) which is fixed and must be used consistently across the organization is stored in the database. The users are able to choose the prepopulated information from dropdown lists. A web form allows to add/edit/delete prepopulated information, but this functionality is restricted to admin users.

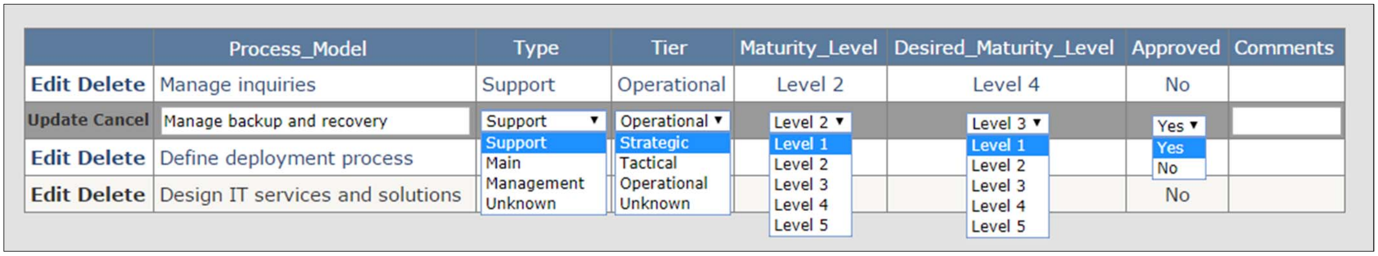


Fig 5. Edit View

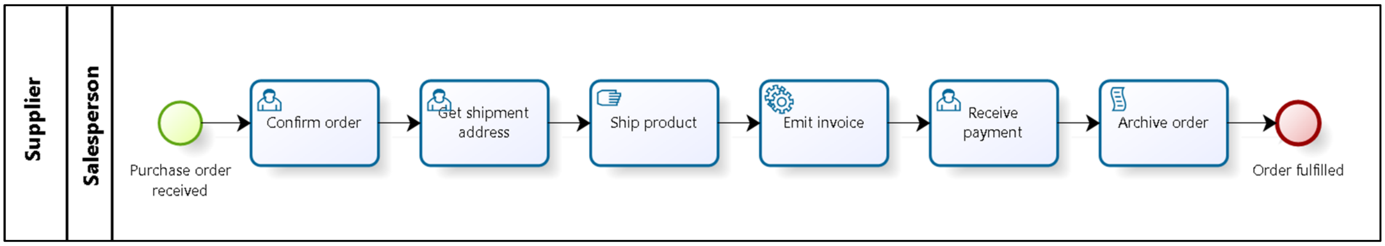


Fig 6. Order fulfillment process - Version 1

IV. Functional Demonstration

In this section, we present various use case scenarios to demonstrate the end-to-end functionality of the proposed framework and its advantages compared with existing tools. First, the process of generating a business process model is demonstrated, which enforces collection of all relevant information to build accurate and BPMN complaint process models. Then the functionality of modifying existing business process model is presented. Next, the ability to align business processes to organization SBOs is described which helps in integration and standardization of processes across all organizational units. Tagging business processes with a maturity level assists the organization to know the current state of each process and provides an opportunity for managers and decision makers to improve underperforming processes. Finally, attaching process tier/type relates processes to the right accountability level and helps in the prioritization of work.

A. Generating business process model

To demonstrate how a business process description is transformed into BPMN model by our proposed framework, we use an order fulfillment process, taken from [1], which is described as: "The process starts whenever a purchase order is received from a customer. The first activity that is carried out is confirming the order. Next, the shipment address is received followed by emitting the invoice; once the payment is received the order is archived, thus completing the process." As an initial step, all information must be gathered in accordance with the specified prerequisite instructions. This consists of *events* (purchase order received and order fulfilled), and *activities* (confirm order, get shipment address, ship product, receive payment and archive order). The information about the role (sales person) and department (supplier) is not provided in process description. Therefore, prerequisite steps enforce gathering of such missing information before the process specifications are submitted. Upon successful submission of all required information, an XPDL file is generated and imported into Bizagi Modeler [13] which provides a visual representation of the model (Fig 6). This step has been automated via an extensive suite of code components using C# and ASP.Net framework which integrates with a backend SQL database.

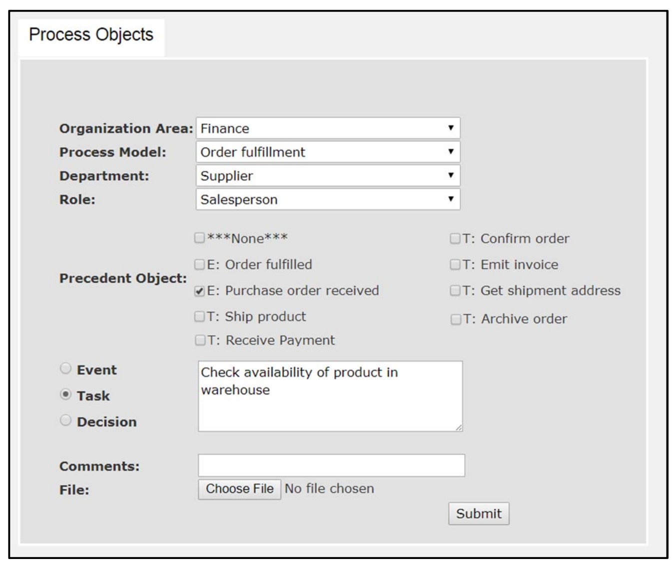


Fig 7. Adding new information to business process model

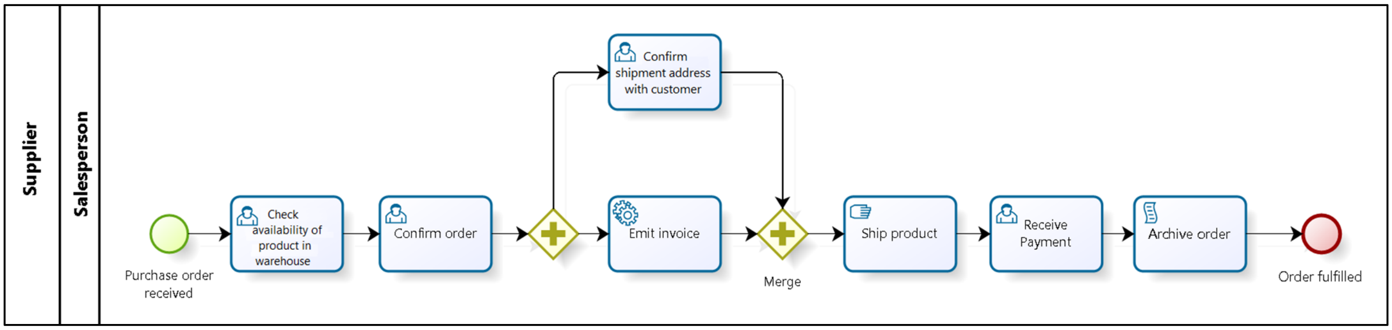


Fig 8. Order fulfillment process - Version 2

B. *Editing existing business process model*

Often, business processes require additions/changes and new models need to be generated. Our proposed framework allows such changes to existing model thereby eliminating the need for a redesign from scratch. To demonstrate this, we extend the order fulfillment process by adding an activity to check availability of product in warehouse before confirming the order. Further, we add an activity to confirm shipment address which can happen in parallel with emitting of the invoice. These changes are added to the model using the process objects form (Fig 7). After submitting the changes, the new business process model is generated as shown in Fig 8. This demonstrates the ability to reuse existing models and avoid redundant work.

C. *Alignment of business processes to SBOs*

Business architecture helps decision makers to better understand the current processes and their relationship with each strategic business objective thereby leading to standardization and optimization of the process. As described by Paul A. Strassmann "Alignment is the capacity to demonstrate a positive relationship between information technologies and the accepted financial measures of performance" [24]. To perform such alignment in our proposed framework, the SBOs need to be linked with CSFs which are in turn linked with functions followed by the specific business processes. For example, order fulfillment process can be aligned to a SBO (Decrease expenses by 5%) by using the forms shown in Fig 9.

Once the alignment information is submitted, linkage between the SBO and business process is displayed in reports (Fig 10). In this example, the SBO (decrease expenses by 5%) is linked to three different CSFs (expand product range to attract more customers, focus on needs & utilizing technologies). One of these CSFs (*expand product range to attract more customers)* is in turn related to two functions (product order & payment and expenses). The *product order* function is finally related to the *order fulfillment* process. In other words, this specific process is aligned with an SBO (decrease expenses) through an intermediate function and a CSF. All reports are dynamic, and changes to the information are instantly reflected in the reports.

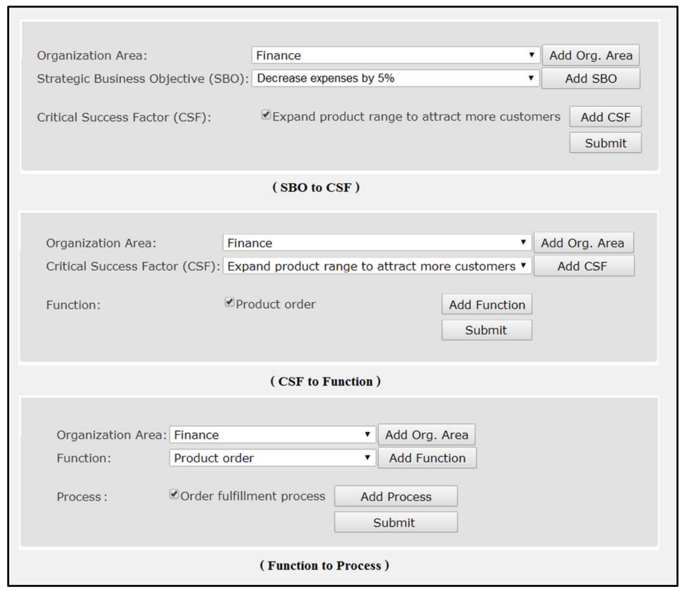


Fig 9. Alignment of SBO to business process forms

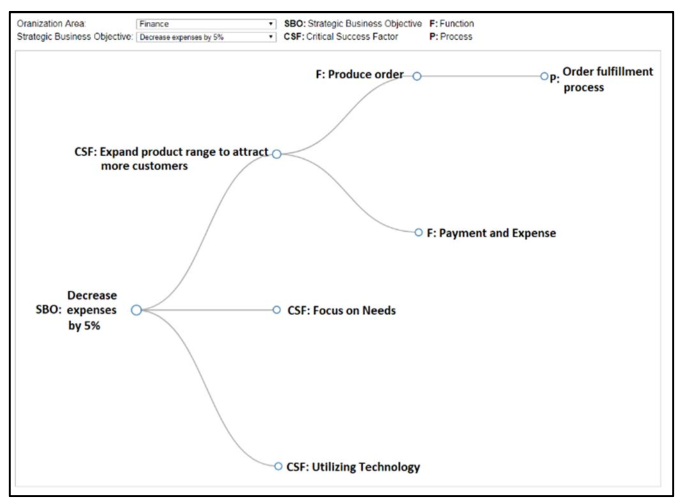


Fig 10. Order fulfillment process alignment to SBO

D. Maturity Level and Process Type/Tier

Tagging of maturity level to a business process assists the organization in knowing the current state of that process. Processes with lower maturity level are candidates for improvement. The current and desired maturity level for each process is captured during submission of process requirements. After successfully improving a process, its maturity level can be updated, which then automatically updates the reports. For example, as shown in Fig 11 the current maturity level of the order fulfillment process is 3 whereas the desired maturity level is 4.

Maturity Level	
# of Process / Reached Desired Maturity Level	10 / 8

Org Area	Process Model	Reached Desired Maturity Level
Information Technology	Manage backup and recovery	✓
	Manage inquiries	✓
	Plan change deployment	✗
	Define deployment process	✓
	Design IT services and solutions	✓
Other	Order fulfillment	✗ Current Maturity: Level 3 Desired Maturity: Level 4
	Order Processing	✓
	Invoice checking	✓
	Bank Account Opening	✓
	Job Offering Process	✓

Fig 11. Report - Maturity level

Finally, a process can also be classified as of a certain type (core, management and support). Similarly, a process can belong to a specified tier (strategic, tactical and operational). This classification assists in relating processes to the right accountability level and thus helps managers and decision makers in prioritization of the work. For example, improving a process which is in operational tier will have less impact than the one in strategic tier as the latter affects the entire organization. Tagging information is captured at the time of process requirement submission or can be added later (Fig 5). The main dashboard includes sub-reports which show process tiers and process types.

V. Conclusion

In this paper, we have demonstrated a framework which not only creates business process models more efficiently and effectively but also aligns the processes with strategic business objectives of the organization. This is accomplished by structured collection of information. The existing traditional procedures used for collecting information to create business process models is slow, expensive and generally leads to misunderstanding or ambiguity between model and domain experts. In our proposed framework, the input obtained is validated at time of data entry, converted into XPDL file format and transformed into BPMN-compliant business process models. The framework allows users to add strategic business objectives, critical success factors, KPIs, functions, and processes and specify the relationship between them. The feedback loop functionality allows users to submit changes to existing models. The constraints and validation are integrated into the framework to minimize redundancy and improve accuracy of the information captured by the framework. The proposed framework also helps to document the processes thus providing for automation and simplified maintenance.

An intuitive dashboard displays aggregated information with capability to navigate across reports and drill down to finer granularity. Among other things, these reports show the maturity level of functions and processes, process type, process tier, and alignment of processes to SBOs. Any changes made to the process(es) are immediately reflected in the reports.

While the framework is capable of building models using core BPMN artifacts, this work can be extended in various directions. For instance, the business process models generated by our framework can be converted into executable models for automation, monitoring and controlling. Similarly, advanced BPMN elements can be implemented and a running visual model could be presented as the user enters data.

References

[1] M. Dumas, M. L. Rosa, J. Mendling and H. Reijers, Fundamentals of Business Process Management, Berlin: Springer, 2013, pp. 1-96.

[2] K. D. Swenson and M. v. Rosing , "Phase 4: What Is Business Process Management," in *The Complete Business Process Handbook: Body of Knowledge from Process Modeling to BPM*, vol. Volume I, Morgan Kaufmann, 2014, pp. 79-88.

[3] M. Will and V. d. Aalst, "A decade of business process management conferences: personal reflections on a developing discipline," in *Business Process Management*, Springer, 2012, pp. 1-16.

[4] S. Lusk, S. Paley and A. Spanyi, "The evolution of business process management as a professional discipline," *Evolution of BPM as a Professional Discipline,* 2005.

[5] G. Redding, M. Dumas, A. H.M. ter Hofstede and A. Iordachescu, "Generating Business Process Models from Object Behavior Models," *Information Systems Management,* vol. 25, pp. 319-331, 2008.

[6] F. Friedrich, J. Mendling and F. Puhlmann, "Process model generation from natural language text," in *Advanced Information Systems Engineering*, Berlin, Springer, 2011, pp. 482-496.

[7] D. Lubke and K. Schneider, "Visualizing use case sets as bpmn processes," in *Requirements Engineering Visualization*, IEEE, 2008, pp. 21-25.

[8] J. M. Kuster, K. Ryndina and H. Gall, "Generation of business process models for object life cycle compliance," in *International Conference on Business Process Management*, Springer, 2007, pp. 165-181.

[9] C. Fritz, R. Hull and J. Su, "Automatic construction of simple artifact-based business processes," in *Proceedings of the 12th International Conference on Database Theory*, ACM, 2009, pp. 225-238.

[10] iGrafx, "Flowcharter," iGrafx, 2016. [Online]. Available: http://www.igrafx.com/products/process-modeling-analysis/flowcharter. [Accessed 29 08 2017].

[11] Visual-Paradigm, "Business Process Modeling Tools (BPMN)," Visual Paradigm, [Online]. Available: https://www.visual-paradigm.com/features/business-process-modeling/.

[12] Oracle, "Oracle Business Process Management Suite 12c," Oracle, [Online]. Available: http://www.oracle.com/us/technologies/bpm/suite/overview/index.html.

[13] Bizagi, "The Digital Business Platform," Bizagi, 2019. [Online]. Available: http://www.bizagi.com/en/products. [Accessed 07 05 2019].

[14] IBM, "Reinvent business processes that are instant, seamless and insightful- IBM Business Process Manager," IBM, [Online]. Available: http://www-03.ibm.com/software/products/en/business-process-manager-standard. [Accessed 21 04 2017].

[15] Bonitasoft, "Bonita BPM," [Online]. Available: https://www.bonitasoft.com/business-process-management-bpm. [Accessed 01 05 2019].

[16] A. Meyer and M. Weske, "Extracting data objects and their states from process models," in *Enterprise Distributed Object Computing Conference (EDOC), 2013 17th IEEE International*, IEEE, 2013, pp. 27-36.

[17] T. Skersys, R. Butleris, K. Kapocius and T. Vileiniskis, "An Approach for Extracting Business Vocabularies from Business Process Models," *Information technology and control,* vol. 42, pp. 150-158, 2013.

[18] H. Leopold, J. Mendling and A. Polyvyanyy, "Supporting process model validation through natural language generation," *IEEE Transactions on Software Engineering,* vol. 40, pp. 818--840, 2014.

[19] I. Kitzmann, C. Koni, D. Lubke and L. Singe, "A simple algorithm for automatic layout of bpmn processes," in *Commerce and Enterprise Computing, 2009. CEC'09. IEEE Conference on*, IEEE, 2009, pp. 391--398.

[20] H. Van Der Aa, H. Leopold and H. A. Reijers, "Detecting Inconsistencies Between Process Models and Textual Descriptions," in *Business Process Management*, Springer, 2015, pp. 90--105.

[21] B. Wetzstein, Z. Ma, A. Filipowska, M. Kaczmarek, S. Bhiri, S. Losada, J.-M. Lopez-Cobo and L. Cicurel, "Semantic Business Process Management: A Lifecycle Based Requirements Analysis," in *Proceedings of the Workshop on Semantic Business Process and Product Lifecycle Management (SBPM 2007)*, vol. 251, 2007, pp. 1-11.

[22] R. Dijkman, M. Dumas, B. v. Dongen, R. Kaarik and J. Mendling, "Similarity of business process models: Metrics and evaluation," *Information Systems,* vol. 36, pp. 498-516, 2011.

[23] F. Pittke, P. H. Piccoli Richetti, J. Mendling and F. Araujo Bai˜ao, "Context-sensitive textual recommendations for incomplete process model elements," in *International Conference on Business Process Management*, Springer, 2015, pp. 189-197.

[24] P. A. Strassmann, "What is alignment? Alignment is the delivery of the required results," Cutter IT Journal, August 1989. [Online]. Available: http://www.strassmann.com/pubs/alignment/. [Accessed 27 04 2017].